$\mathcal{O}(n)$ alternative to Quantum Fourier Transform with efficient neural net classical post-processing

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The Quantum Fourier Transform (QFT) is required by hidden subgroup problem (HSP) algorithms, including Shor's algorithm for factoring. The circuit depth of the QFT remains challenging for near-term hardware. To find shallower alternatives we identify two properties that are exploited by the QFT to enable HSP. Firstly, the shift invariance of the QFT allows for the removal of a random overall shift. Secondly, the QFT retains information about the hidden subgroup generator accessible in the measurement outcomes. We quantify that information via the discrete Fisher information. We construct a family of shallow circuits using Hadamards and controlled-Phase gates, HP-L circuits, and prove shift invariance for the fixed-phase HP-1 instance via a general phase-factorization criterion. Numerical analysis shows these circuits retain exponentially growing Fisher information. The $\mathcal{O}(n)$ HP-1 can replace the $\mathcal{O}(n^2)$ QFT in Shor's algorithm, as demonstrated numerically, with an efficient neural network implementing classical post-processing.

Introduction.—The $\mathcal{O}(n^2)$ circuit depth required by the standard Quantum Fourier Transform (QFT) presents experimental challenges. The QFT is used in the exponential quantum speedups of landmark protocols—such as Shor's algorithms for prime factorization and discrete logarithms [1–3]. The associated family of algorithms, all using the QFT, is known as the (quantum) hidden subgroup problem (HSP) algorithms [4–7]. The experimental implementation of the QFT remains challenging. For a 100-qubit system ($n = 100$), the QFT demands thousands of circuit layers, which vastly exceeds current hardware capabilities. For instance, IBM's 127-qubit Eagle processor is limited to a depth of 60 layers even with extensive error mitigation [8]; Google's 105-qubit Willow processor reaches its operational threshold at around 40 layers [9]; USTC's 105-qubit Zuchongzhi 3.0 processor demonstrated random circuit sampling on 83-qubits, with around 32 layers [10]. Moreover, for quantum error-mitigation protocols that are commonly used, the sampling overhead can scale exponentially with the circuit depth [11, 12].

There has been significant progress in reducing the circuit depth required. These efforts have enabled record experimental QFT fidelities [13] through techniques including (i) approximate or truncated QFT constructions, which omit small-angle controlled rotations [14, 15] with $\mathcal{O}(n \log n)$ depth, (ii) dynamic mid-circuit measurements, which reduce the resource scaling of QFT followed immediately by measurement from $\mathcal{O}(n^2)$ two-qubit gates to $\mathcal{O}(n)$ mid-circuit measurements without connectivity constraints [16], and (iii) quantum circuit cutting and cluster-simulation methods, which trade coherent quantum depth for additional sampling and classical post-processing overhead [17, 18].

We here take a different approach to reducing the circuit depth required. Rather than approximate the QFT we identify key features of the QFT that make it compatible with the HSP. We focus on shift-invariance combined with retention of information about the generator in question. Fig. 1(a) summarizes the main idea. We design non-QFT circuits that retain those key features.

We construct a circuit family, called HP-L circuits. In particular, the HP-1 circuits are shallow and retain information about the generator. The HP-L circuits interleave transversal Hadamard (H) gates with L layers of phase (P) blocks. We give a general algebraic criterion for shift invariance and verify it analytically for the fixed-phase HP-1 circuit used below. Numerical scaling analysis demonstrates exponentially increasing retention of Fisher information $\sim e^{kn}$, albeit with a worse exponent k than QFT. That $\sim e^{kn}$-scaling implies efficient scaling of the number of samples required.

We numerically simulate replacing the QFT in Shor's algorithm with HP-1 circuits, and using an efficiently scaling neural network [19] for classical post-processing. Factorisation is successfully achieved.

Necessary condition for Shift Invariance.—The standard quantum HSP algorithm is illustrated in Fig. 1(b) with gray blocks [1–3, 5]. After U_f, the state is proportional to $\sum_x |x\rangle |f(x)\rangle$. The elements x belong to a finite abelian group G with group operation $+$. We are guaranteed that $f(x + r) = f(x)$ for the unknown generator r that we wish to find. In other words, the function $f(\cdot)$ is constant on cosets of V, the subgroup generated by r. Upon measuring (or imagining measuring) the second register to find $|f(c)\rangle$ the state becomes proportional to $\sum_{q\in\{0,1,2...\}} |c + qr\rangle |f(c)\rangle$ with the shift c acting as undesired noise obfuscating r (additional details about the

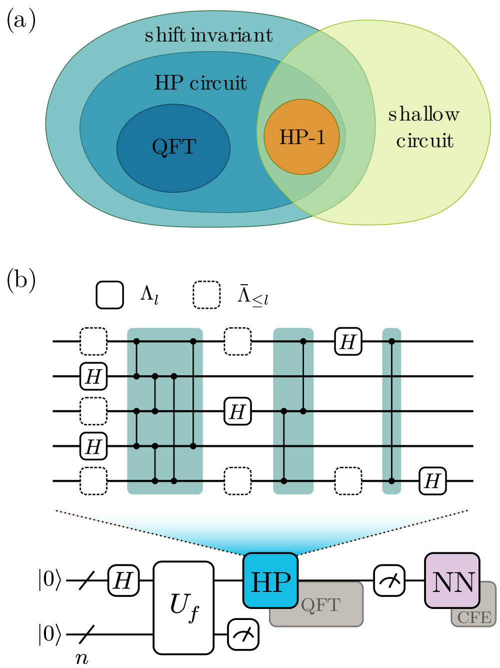

FIG. 1. **Shallow shift-invariant alternative to the QFT.** (a) Shift invariance is a critical property of the QFT. We introduce a circuit family, denoted as HP-L, and prove this property for the fixed-phase HP-1 circuit used below. Within this broader family, HP-1 specifically serves as a shallow-depth circuit that is experiment-friendly. (b) In the proposed HSP algorithmic routine, the deep QFT block (the left gray gate) is directly replaced by the HP circuit. A neural network then processes the measurement outcomes, bypassing the standard continued fraction expansion (CFE) typically used in the standard HSP. The HP circuit is composed of Hadamard and phase gates, with Λ_l referring to the qubits with Hadamards applied in layer l.

HSP algorithm are provided in Appendix A).

We now define shift invariance [20] for HSP sampling and derive a necessary condition that any shift-invariant unitary must satisfy. For a subgroup $V \leq G$, in particular $V = \{|qr\rangle\}$, we denote the uniform superposition over its elements as $|V\rangle := \sum_{q\in\{0,1,2\dots\}} |qr\rangle = \frac{1}{\sqrt{|V|}} \sum_{v\in V} |v\rangle$, for notational simplicity.

Definition 1 (Shift invariance)**.** *A unitary U acting on the group register is shift-invariant for HSP sampling if*

$$|\langle x| U |V\rangle|^2 = |\langle x| U |c+V\rangle|^2, \quad \forall x, c \in G,\ \forall V \leq G. \quad (1)$$

To identify shift-invariant circuits beyond the QFT, we first establish a necessary condition for shift invariance.

Proposition 1 (Necessary condition of shift invariance)**.** *Suppose U is a unitary on the group register that satisfies the shift-invariance condition above. Then, for every $x, y \in G$ there exist phases $\theta_{xy} \in \mathbb{R}$ such that*

$$\langle x| U |y\rangle = \frac{1}{\sqrt{|G|}} e^{\mathrm{i}\theta_{xy}}. \quad (2)$$

Proof sketch of Proposition 1. The proof hinges on analyzing the shift-invariance condition for the trivial subgroup $V = \{0\}$. For the subgroup $V = \{0\}$, the state $|V\rangle$ is the computational basis state $|0\rangle$, and the shifted state is $|c+V\rangle = |c\rangle$. The requirement (Eq.(1)) that $|\langle x| U |0\rangle|^2 = |\langle x| U |c\rangle|^2$ for all $x, c \in G$ implies that, for any fixed output x, the magnitude of every matrix element in the x-th row of U must be identical. The special case induces the uniform magnitude $\langle x| U |y\rangle = 1/\sqrt{|G|}e^{\mathrm{i}\theta_{xy}}$. The full details of this argument are provided in Appendix B. □

Hadamard-Phase circuits and fixed-phase HP-1 shift invariance.—Proposition 1 suggests a simple design principle for constructing shift-invariant circuits. The condition $|\langle x|U|y\rangle| = 1/\sqrt{|G|}$ requires flat amplitude magnitudes, which are naturally generated by Hadamard gates. The remaining freedom lies in the relative phases θ_{xy}, which can be adjusted by controlled-phase gates ($\mathrm{CP}(\theta) = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + e^{\mathrm{i}\theta}|11\rangle\langle 11|$) without changing these magnitudes. This motivates the Hadamard-phase (HP) circuit family we shall now introduce. As illustrated in Fig. 1 (b), the HP-L circuit is structured as an alternating sequence of Hadamard gates and phase blocks. The architecture is chosen so that the resulting phases can be checked against the shift-invariance criterion below: (i) each qubit is acted on by a Hadamard gate exactly once; (ii) each qubit pair is connected by at most one controlled-phase gate; and (iii) each controlled-phase gate must come together with 2 Hadamards, one before and one after the phase gate. Here, "once" refers to the entire L-layer circuit rather than to a single layer.

To express the allowed circuits mathematically, we define two types of sets of qubits. Let H_i denote the Hadamard gate applied to qubit $i \in \{1, 2, \dots, n\}$. For each layer l, let Λ_l denote the set of qubits on which Hadamard gates are applied in that layer. The qubits that are still allowed in terms of having a Hadamard on them after layer l, i.e. those that have not had a Hadamard on them in or before layer l, are denoted as the set $\bar{\Lambda}_{\leq l}$.

In-between the Hadamard layers, the two sets are coupled via a controlled-phase block, denoted as $\mathrm{CP}_{\Lambda_l,\bar{\Lambda}_{\leq l}}(\boldsymbol{\theta}_{\Lambda_l\bar{\Lambda}_{\leq l}})$, which applies pairwise operations $\mathrm{CP}_{jk}(\theta_{jk})$ between each qubit $j \in \Lambda_l$ and $k \in \bar{\Lambda}_{\leq l}$. Building upon this notation, we formally define the circuit architecture:

Definition 2 (HP-L circuit)**.** *A quantum circuit is classified as an HP-L circuit if it is constructed using the* *following layer-wise architecture*

$$U_{\mathrm{HP}}^{(L)}(\boldsymbol{\theta}) = \bigotimes_{i\in\Lambda_L} H_i \prod_{l=L-1}^{1} \left(\mathrm{CP}_{\Lambda_l,\bar{\Lambda}_{\leq l}}(\boldsymbol{\theta}_{\Lambda_l\bar{\Lambda}_{\leq l}}) \bigotimes_{i\in\Lambda_l} H_i \right), \tag{3}$$

where the controlled-phase block is given by

$$\mathrm{CP}_{\Lambda_l,\bar{\Lambda}_{\leq l}}(\boldsymbol{\theta}_{\Lambda_l\bar{\Lambda}_{\leq l}}) = \prod_{j\in\bar{\Lambda}_{\leq l}} \prod_{k\in\Lambda_l} \mathrm{CP}_{jk}(\theta_{jk}). \tag{4}$$

The parameter set $\boldsymbol{\theta}_{\Lambda_l\bar{\Lambda}_{\leq l}}$ is the collection of θ_{jk}, $j \in \Lambda_l$, $k \in \bar{\Lambda}_{\leq l}$.

The shift-invariance proof is most transparent when separated into a general algebraic observation and the specific circuit used below.

Lemma 1 (Coset shifts as removable phases)**.** *Let $G = \mathbb{Z}_{2^n}$, encoded by n computational-basis bits. Suppose a unitary U has matrix elements of the form*

$$\langle x| U |y\rangle = \frac{1}{\sqrt{2^n}} \exp\left[\mathrm{i} \sum_{i,j=1}^{n} A_{ij} x_i y_j \right], \qquad A_{ij} \in \mathbb{R}. \tag{5}$$

Then U is shift-invariant for HSP sampling over $\mathbb{Z}_{2^n}$.

For the main circuit used in the numerical sections, we take the fixed-phase HP-1 skeleton $\Lambda_1 = \{1, 3, 5, \ldots\}$ and $\Lambda_2 = \{2, 4, 6, \ldots\}$, with $\theta_{ij} = 2\pi/2^{|i-j|}$ on each controlled-phase gate between $i \in \Lambda_1$ and $j \in \Lambda_2$.

Theorem 2 (Shift invariance of fixed-phase HP-1)**.** *The fixed-phase HP-1 circuit is shift-invariant for HSP sampling over $\mathbb{Z}_{2^n}$.*

The proof verifies that the fixed-phase HP-1 matrix elements have the bilinear phase form required by Lemma 1 (Appendix C).

HP-0 can be used in HSP over $\mathbb{Z}_{2^n}$.—We give an analytical solution for HSP over $\mathbb{Z}_{2^n}$ via parallel Hadamard gates. We consider the cyclic group $\mathbb{Z}_{2^n}$ since the group can be naturally represented in the computational basis. In this case, each subgroup is generated by a power of two,

$$V = \{q2^p : q = 0, \ldots, 2^{n-p} - 1\} \leq \mathbb{Z}_{2^n}, \tag{6}$$

where p specifies the subgroup generator 2^p. Applying a transversal Hadamard layer to a coset state gives

$$H^{\otimes n} |c + V\rangle \propto \sum_{k\in\{0,1\}^p} (-1)^{k\cdot c} |0\rangle^{\otimes(n-p)} |k\rangle. \tag{7}$$

By measuring the state $H^{\otimes n} |c + V\rangle$, one observes that the first $n-p$ bits are always fixed to 0, while the remaining p bits are uniformly distributed over $\{0, 1\}^p$. Thus, the value of p can be determined by identifying which bit

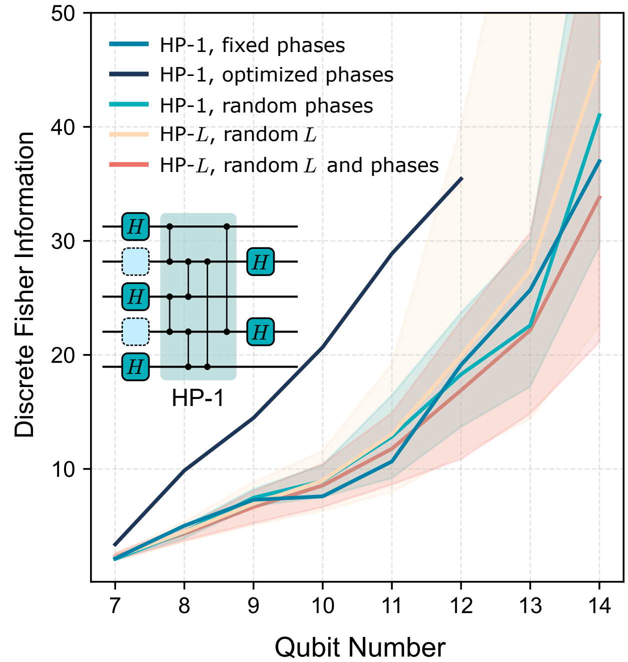

FIG. 2. **Average discrete Fisher information (DFI) increasing with the number of qubits.** Results are shown for several HP-family circuit settings. L denotes the number of phase blocks layers with HP-1 having a single phase block layer. Solid curves show the mean, with shaded bands indicating one standard deviation associated with random phase choices.

positions are consistently zero. The detailed calculation and sampling algorithm are provided in Appendix D.

HP-L for HSP over $\mathbb{Z}$: information on generator is retained.—The protocol HP-0 fails for general cyclic groups $\mathbb{Z}_q$ because the measurement distributions corresponding to different subgroups V are not distinguishable. That absence of distinguishability demonstrates that shift invariance alone is insufficient for successful hidden subgroup recovery, and that distinguishability should be imposed as an additional condition.

We use the discrete Fisher information (DFI) as a metric for distinguishability.

$$\mathrm{DFI}(r, n) = \sum_{x=0}^{2^n-1} \frac{(\Pr(x \mid r+1) - \Pr(x \mid r))^2}{\Pr(x \mid r)}, \tag{8}$$

where x represents measurement outcomes, r is the hidden subgroup generator, and $\Pr(x \mid r)$ is the probability of measuring x when the subgroup generator is r. The sample complexity for estimating a discrete parameter is formally governed by the DFI (Eq.(8)) via the Hammersley-Chapman-Robbins bound [21, 22],

$$m \geq \frac{\ln\left(1 + \epsilon^{-2}\right)}{\ln\left(1 + \mathrm{DFI}(r)\right)}, \tag{9}$$

where m is the required number of samples and ϵ denotes the maximum tolerable standard deviation of the estimator of the parameter. The bound suggests that retaining an exponentially growing DFI results in tractable (m lower-bounded by $o(\frac{1}{n})$) sample complexity for the subsequent neural-network-based post-processing.

For the standard QFT (Appendix E), we can calculate the DFI as

$$\mathrm{DFI}_{\mathrm{QFT}}(r,n) \approx \frac{4\pi^2}{9}\left(\frac{2^{2n}}{r^2}-1\right) \sim \mathcal{O}\left(2^{2n}\right). \tag{10}$$

Eq. (10) indicates that the generator r becomes easier to recover as the qubit number increases, which motivates us to limit r to $1 \ll r \ll 2^n$.

For HP circuits, the DFI is evaluated numerically. Specifically, for a fixed r, the circuit maps the periodic input components $\{|qr\rangle\}_q$ to a computational-basis distribution over outcomes $|b\rangle$, with the exact probabilities given by $\Pr(|b\rangle) = \left|\sum_q \langle b|U|qr\rangle\right|^2$. To evaluate the worst-case parameter sensitivity as the system scales, we consider the worst-case DFI: $\mathrm{DFI}_{\min}(n)$: $\min_{1\le r\le \lfloor 2^{n/2}\rfloor} \mathrm{DFI}(r,n)$. Fig. 2 illustrates the minimum DFI against the qubit number n. (We bound the period to $1 \le r \le \lfloor 2^{n/2} \rfloor$ to satisfy the condition $r \ll 2^n$.)

Fig. 2 also compares the scaling of different HP circuit configurations, where solid curves and shaded bands represent the mean and standard deviation of $\mathrm{DFI}_{\min}(n)$. The fixed-phase HP-1 circuit utilizes the skeleton, $\Lambda_1 = \{1,3,5,\cdots\}$ and $\Lambda_2 = \{2,4,6,\cdots\}$, with controlled-phase gates with phase $\frac{2\pi}{2^{|i-j|}}$ between qubits i and j. The optimized HP-1 circuit adopts the same skeleton but treats the gate phases as variational parameters, trained to maximize the $\mathrm{DFI}_{\min}$. The deeper HP-L circuits are constructed by generating valid random skeletons Λ.

The proposed HP circuit family preserves an exponential growth of DFI, based on the numerical evidence. As detailed in Appendix F, numerical data for both the HP circuits and the QFT benchmark fit the log-linear model $\ln \mathrm{DFI}_{\min}(n) = kn+b$, with a coefficient of determination $R^2 \ge 0.913$.

Efficient neural network for classical post-processing.—To avoid the exponential costs associated with processing full probability distributions, we employ a poly(n) Deep Sets neural network [19, 23] to predict the hidden subgroup generator directly from the raw measurement outcomes $\{b_i\}$. The input data is generated using the optimized HP-1 circuit evaluated in Fig. 2.

The outcomes are fed into a shared feature map ϕ, which maps each bitstring b_i to a feature vector within 2 linear layers and $16n$ feature dimension. The feature vectors are then summed to form an aggregated, permutation-invariant representation $\sum_i \phi(b_i)$. Subsequently, a beam-search decoder [24, 25] performs a top-k search to rank candidate periods. We use polynomial samples $m = 1024n^2$ to train the neural net. This architecture avoids exponential sample-space computations and outputs a ranked list of candidate generators for classical verification.

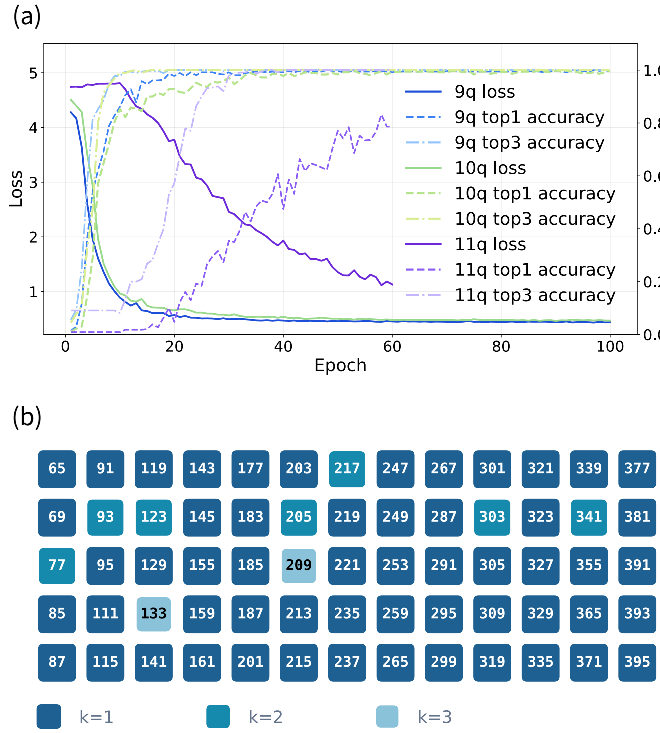

FIG. 3. **Numerical simulation shows successful Shor factoring via HP-L and neural net post-processing.** There is a depth-7 HP-1 circuit with only 11 Hadamards and 30 controlled-phase gates on 11 qubits. (a) Training curves for 9-, 10-, and 11-qubit instances show rapid improvement in generator recovery accuracy. (b) Shor-type factorization of numbers ≤ 395 is demonstrated. Colour indicates the minimum number k of HP-1-based guesses needed for successful factorization. Most instances are solved with $k = 1$, while only a few require $k = 2$ or $k = 3$.

Replacing $\mathcal{O}(n^2)$ QFT with $\mathcal{O}(n)$HP-1 in Shor's Algorithm.—We evaluate the optimized HP-1 circuit in classical numerical simulations of Shor's algorithm for factoring with neural net classical post-processing. Shor's algorithm seeks the generator r satisfying $a^r \equiv 1 \pmod N$. We select the base a such that $1 \le r \le 2^{n/2}$, ensuring resolvability by the n-qubit register. We consider factoring of numbers up to 395.

The neural net trained successfully. As shown in Fig. 3(a), the neural network achieves rapid convergence when trained on 9-, 10-, and 11-qubit instances. Both the top-1 (correct period ranked first) and top-3 accuracies increase swiftly within a few epochs.

The combined HP-1 and neural net protocol recovers the subgroup generator in the tested instances. As illustrated in Fig. 3(b), each composite integer is colored by the minimum number of guesses k. Most instances are solved on the first attempt ($k = 1$) and rest within k=3.

Noise simulation under a post-circuit global depolarizing channel demonstrates that our protocol with the optimized HP-1 circuit is robust against moderate environmental noise. The recovery accuracy remains stable up to noise-strength $\eta \approx 0.215$ for 10 qubits ($1-\eta$ is the probability of applying the identity to the input state). Appendix G provides detailed settings, complete numerical results, and corresponding interpretations.

Summary and outlook.—We propose a hardware-efficient framework for solving the HSP by shallow, shift-invariant HP-1 circuits with a polynomial-scaling neural network. To verify the scalability of this protocol, we analyzed the growth model of the DFI. Finally, we evaluated our protocol on Shor's algorithm. The numerical simulations demonstrate that our approach successfully finds the hidden subgroup with high accuracy and remains robust against moderate depolarizing noise.

Realizing a practical HSP protocol further requires optimized implementations of the oracle U_f. Related strategies for reducing oracle and data-loading costs include Hamiltonian-learning-based state preparation that transfers the cost to classical preprocessing and uses $\mathcal{O}(1)$ oracle queries [26], space-time tradeoffs for non-Clifford gate reductions [27], and automatic oracle synthesis frameworks [28].

Apart from the $\mathcal{O}(n)$ scaling being hardware friendly, the reduction to grouped phase gates in HP circuits makes the phase operations easier to implement than individual phase gate execution. This hardware-level efficiency arises because consecutive phase operations can be merged into single continuous pulses across multiple experimental platforms, including in linear optics [29, 30], neutral-atom systems [31–33], trapped ions [34, 35], and NMR [36].

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Code Availability.—All code relevant to this work is publicly available in the GitHub repository: `https://github.com/Fragecity/altqft`.

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APPENDIX

A. Hidden Subgroup Problem and Fourier Sampling

For completeness, this section reviews the standard quantum algorithm for solving the Hidden Subgroup Problem (HSP) over finite abelian groups [4–6]. Formally, the HSP is defined as follows:

Definition 3 (Hidden Subgroup Problem)**.** *Given a function $f : G \to X$ defined on a group G, with the promise that there exists a hidden subgroup $V \leq G$ such that for all $g_1, g_2 \in G$*

$$f(g_1) = f(g_2) \iff g_1 - g_2 \in V \tag{A1}$$

The objective of this task is to identify the hidden subgroup V.

In the context of this paper, we restrict our attention to finite abelian groups. For such groups, the fundamental theorem of finite abelian groups allows us to express the group structure as a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{p_1} \times \mathbb{Z}_{p_2} \times \cdots \times \mathbb{Z}_{p_k}. \tag{A2}$$

The standard quantum procedure for solving the hidden subgroup problem begins by preparing a uniform superposition over all elements of the group G [4, 20]. We initialize the system in the state

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |0\rangle , \tag{A3}$$

where $|G|$ is the number of elements of the group G. The first register holds the group elements, and the second register is initialized to a standard basis state. Next, we evaluate the quantum oracle corresponding to the function

f, which acts on the two registers and entangles the group elements with their corresponding function values

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |f(g)\rangle . \tag{A4}$$

Grouping the superposition terms according to cosets is illuminating, as the function f is constant within each coset of the hidden subgroup V and distinct across different cosets.

Before proceeding, it is useful to formalize the state notation for a subset. For any subset $S \subseteq G$, we define its corresponding normalized quantum state $|S\rangle$ as the uniform superposition over all elements within the set:

$$|S\rangle = \frac{1}{\sqrt{|S|}} \sum_{s \in S} |s\rangle . \tag{A5}$$

Using this definition, the normalized state for a specific coset $c + V$ is directly given by

$$|c + V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |c + v\rangle . \tag{A6}$$

By applying this coset state definition, the post-oracle state can be elegantly rewritten as a superposition of these orthogonal coset states, each perfectly entangled with its unique function evaluation:

$$|\psi\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{c \in C} |c + V\rangle |f(c)\rangle , \tag{A7}$$

where C is the set of all possible coset representatives. For simplicity, we assume that the second register is measured immediately after the oracle query. Measuring the second register yields a specific outcome $f(c)$ with a uniform probability of $|V|/|G|$. Consequently, the first register collapses into the corresponding pure coset state:

$$|\psi_{\text{post}}\rangle = |c + V\rangle . \tag{A8}$$

From this point forward, our analysis will focus exclusively on the quantum state of this first register.

Whether the second register is physically measured to collapse the first register into a pure coset state $|c + V\rangle$, or traced out to leave it in a mixed state, the resulting Fourier-basis statistics remain identical [20]. The quantum Fourier transform produces a measurement distribution that is independent of the coset representative. For simplicity, we assume the second register is measured immediately after the oracle query, allowing our subsequent analysis to focus exclusively on the pure coset state:

$$|\psi_{\text{post}}\rangle = |c + V\rangle . \tag{A9}$$

A direct measurement in the computational basis is uninformative. Indeed, using the definition of the coset state,

$$\langle g|c + V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} \langle g|c + v\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} \delta_{g,c+v}. \tag{A10}$$

For fixed c, the map $v \mapsto c + v$ is a bijection from V to the coset $c + V$. Therefore the sum $\sum_{v \in V} \delta_{g,c+v}$ equals 1 when $g \in c + V$ and equals 0 when $g \notin c + V$. Hence

$$\Pr[g \mid c, V] = |\langle g|c + V\rangle|^2 = \begin{cases} 1/|V|, & g \in c + V, \\ 0, & g \notin c + V, \end{cases} \tag{A11}$$

so the observed element is uniform on the unknown coset. Averaging over uniformly random cosets gives the marginal distribution

$$\Pr[g] = \frac{1}{|G|}, \tag{A12}$$

which contains no direct information about the hidden subgroup V.

To extract meaningful information, we must apply the QFT over the group G before measuring [4, 5]. Let $\chi_y(g)$ denote the group characters. The QFT maps a computational basis state $|g\rangle$ as follows:

$$\mathrm{QFT}\,|g\rangle = \frac{1}{\sqrt{|G|}} \sum_{y\in G} \chi_y(g)\,|y\rangle\,. \tag{A13}$$

Applying this transformation to our pure coset state $|c+V\rangle$ yields:

$$\begin{aligned}\mathrm{QFT}\,|c+V\rangle &= \frac{1}{\sqrt{|V|}} \sum_{h\in V} \mathrm{QFT}\,|c+v\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{v\in V}\sum_{y\in G} \chi_y(c+v)\,|y\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{y\in G} \chi_y(c)\left(\sum_{v\in V}\chi_y(v)\right)|y\rangle\,. \end{aligned}\tag{A14}$$

By the orthogonality of group characters, the sum $\sum_{v\in V}\chi_y(v)$ evaluates to $|V|$ if the character χ_y is trivial on V (i.e., y belongs to the orthogonal subgroup $V^\perp$), and evaluates to 0 otherwise. Thus, the state simplifies to a superposition over $V^\perp$:

$$\mathrm{QFT}\,|c+V\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{y\in V^\perp} \chi_y(c)\,|y\rangle\,. \tag{A15}$$

Measuring this state in the computational basis gives the probability of observing an outcome $y \in V^\perp$:

$$\Pr[y] = \left|\sqrt{\frac{|V|}{|G|}}\chi_y(c)\right|^2 = \frac{|V|}{|G|}|\chi_y(c)|^2 = \frac{|V|}{|G|}. \tag{A16}$$

Crucially, because $|\chi_y(c)|^2 = 1$ for all group characters, the resulting measurement distribution is uniformly distributed over $V^\perp$ and is strictly independent of the random coset representative c. This shift invariance is the fundamental property that allows the extraction of subgroup information by effectively removing the unknown shift at the probability level.

The QFT over G is defined using the character group $\hat{G}$ by

$$F_G|g\rangle = \frac{1}{\sqrt{|G|}} \sum_{\chi\in\hat{G}} \chi(g)\,|\chi\rangle, \tag{A17}$$

where $\{|\chi\rangle : \chi \in \hat{G}\}$ is the Fourier basis and each character $\chi : G \to \mathbb{C}^\times$ is a group homomorphism. Here $\mathbb{C}^\times = \mathbb{C}\setminus\{0\}$ denotes the multiplicative group of nonzero complex numbers, so with additive notation on G one has $\chi(g_1+g_2) = \chi(g_1)\chi(g_2)$. For finite abelian G, every character has unit modulus. Applying F_G to a coset state gives

$$\begin{aligned}F_G|c+V\rangle &= \frac{1}{\sqrt{|V|}} \sum_{v\in V} F_G|c+v\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{\chi\in\hat{G}}\sum_{v\in V} \chi(c+v)\,|\chi\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{\chi\in\hat{G}} \chi(c)\left(\sum_{v\in V}\chi(v)\right)|\chi\rangle. \end{aligned}\tag{A18}$$

The character sum over V satisfies

$$\sum_{v\in V}\chi(v) = \begin{cases}|V|, & \chi\in V^\perp,\\ 0, & \chi\notin V^\perp,\end{cases} \qquad V^\perp = \{\chi\in\hat{G} : \chi(v) = 1 \text{ for all } v\in V\}. \tag{A19}$$

Hence

$$F_G|c+V\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{\chi \in V^\perp} \chi(c)\,|\chi\rangle. \tag{A20}$$

The Fourier-basis measurement distribution is therefore

$$\Pr(\chi \mid c, V) = \begin{cases} |V|/|G|, & \chi \in V^\perp, \\ 0, & \chi \notin V^\perp, \end{cases} \tag{A21}$$

which is independent of the coset representative c. The same formula holds for the mixed state ρ_V, since it is a convex combination of pure coset states with identical Fourier statistics.

Repeating the procedure therefore produces samples from $V^\perp$. In the finite abelian setting, classical post-processing reconstructs $V^\perp$ from the sample span and then recovers the hidden subgroup using the duality identity [4, 6, 37]

$$(V^\perp)^\perp = V. \tag{A22}$$

Fourier sampling thus converts hidden translational structure in the oracle into explicit linear constraints in the dual group.

B. Necessary condition of shift invariance

Here we make explicit the probability-level shift-invariance condition used in Fourier sampling. Let G be a finite abelian group, and let $\{|g\rangle : g \in G\}$ denote the computational basis. For $c \in G$, define the shift operator

$$T_c\,|g\rangle = |g+c\rangle\,. \tag{B1}$$

For a subgroup $V \leq G$, the normalized subgroup state is

$$|c+V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |c+v\rangle\,. \tag{B2}$$

For HSP sampling, shift invariance requires that the measurement distribution after applying U is strictly independent of the coset representative:

$$|\langle x|\,U\,|V\rangle|^2 = |\langle x|\,UT_c\,|V\rangle|^2\,, \qquad \forall x, c \in G,\ \forall V \leq G. \tag{B3}$$

Operationally, this means $U\,|V\rangle$ and $U\,|c+V\rangle$ may differ by a relative phase, but they must yield identical probabilities in the computational basis. Any viable QFT replacement must therefore satisfy this condition.

Proposition (Necessary condition of shift invariance)**.** *Suppose U is a unitary on the group register that satisfies Eq.* (B3)*. Then, for every $x, y \in G$,*

$$|\langle x|\,U\,|y\rangle| = \frac{1}{\sqrt{|G|}}. \tag{B4}$$

Equivalently, there exist phases $\theta_{xy} \in \mathbb{R}$ such that

$$\langle x|\,U\,|y\rangle = \frac{1}{\sqrt{|G|}} e^{\mathrm{i}\theta_{xy}}. \tag{B5}$$

To determine how this restricts the matrix elements of U, we can examine the trivial subgroup $V = \{0\}$. In this case, the subgroup state is simply the basis state $|0\rangle$, meaning the shift-invariance condition directly constrains the columns of U. Since the output probabilities for $|0\rangle$ and any shift $|g\rangle$ must be equal, for any fixed outcome x, all entries in the x-th row of U must share the same modulus. Unitarity then forces this common modulus to be exactly $1/\sqrt{|G|}$. This leads directly to Proposition 1.

Proof of Proposition 1. Take the trivial subgroup $V = \{0\}$. Then $|V\rangle = |0\rangle$ and $T_c |V\rangle = |c\rangle$ for every $c \in G$. Equation (B3) therefore implies

$$|\langle x| U |0\rangle|^2 = |\langle x| U |c\rangle|^2, \qquad \forall x, c \in G. \tag{B6}$$

Using the matrix element convention $U_{xy} = \langle x| U |y\rangle$, this equality shows that $|U_{x0}|^2 = |U_{xc}|^2$ for all $c \in G$. In other words, for each fixed row x, all entries have the same squared modulus. Denote this common value by a_x. Since U is unitary, the sum of the squared moduli across any row must be normalized to 1:

$$1 = \sum_{y \in G} |U_{xy}|^2 = |G| a_x. \tag{B7}$$

Hence, $a_x = 1/|G|$ for every $x \in G$. Taking the square root yields the uniform modulus for all matrix elements:

$$|U_{xy}| = \frac{1}{\sqrt{|G|}}. \tag{B8}$$

Because this modulus is strictly positive, every entry admits a polar decomposition

$$U_{xy} = \frac{1}{\sqrt{|G|}} e^{\mathrm{i}\theta_{xy}}, \qquad \theta_{xy} \in \mathbb{R}. \tag{B9}$$

This proves the claim. □

C. A phase-factorization lemma and fixed-phase HP-1 shift invariance

This appendix proves Lemma 1 and then applies it to the fixed-phase HP-1 circuit used in the main text. The point of the lemma is to isolate the only structure needed for shift invariance: a coset shift may enter the amplitude, but only as an overall phase for each measurement outcome.

Lemma (Coset shifts as removable phases)**.** *Let $G = \mathbb{Z}_{2^n}$, encoded by n computational-basis bits. Suppose a unitary U has matrix elements of the form*

$$\langle x| U |y\rangle = \frac{1}{\sqrt{2^n}} \exp\left[\mathrm{i} \sum_{i,j=1}^{n} A_{ij} x_i y_j\right], \qquad A_{ij} \in \mathbb{R}. \tag{C1}$$

Then U is shift-invariant for HSP sampling over $\mathbb{Z}_{2^n}$.

Proof. Every subgroup of $\mathbb{Z}_{2^n}$ has the form $V = \langle 2^p \rangle$ for some $0 \leq p \leq n$. With the standard binary ordering used in the main text,

$$|V\rangle = \frac{1}{\sqrt{2^{n-p}}} \sum_{h_1,\ldots,h_{n-p} \in \{0,1\}} |h_1 \cdots h_{n-p} 0 \cdots 0\rangle. \tag{C2}$$

Any coset has a representative $c \in \{0, \ldots, 2^p - 1\}$, so

$$|c + V\rangle = \frac{1}{\sqrt{2^{n-p}}} \sum_{h_1,\ldots,h_{n-p} \in \{0,1\}} |h_1 \cdots h_{n-p} c_{n-p+1} \cdots c_n\rangle. \tag{C3}$$

For a fixed measurement outcome x, define $a_j(x) := \sum_{i=1}^{n} A_{ij} x_i$. Then

$$\begin{aligned} \langle x| U |c + V\rangle &= \frac{1}{\sqrt{2^n 2^{n-p}}} \sum_{h_1,\ldots,h_{n-p}} \exp\left[\mathrm{i} \sum_{j=1}^{n-p} a_j(x) h_j + \mathrm{i} \sum_{j=n-p+1}^{n} a_j(x) c_j\right] \\ &= \exp\left[\mathrm{i} \sum_{j=n-p+1}^{n} a_j(x) c_j\right] \frac{1}{\sqrt{2^n 2^{n-p}}} \sum_{h_1,\ldots,h_{n-p}} \exp\left[\mathrm{i} \sum_{j=1}^{n-p} a_j(x) h_j\right]. \end{aligned} \tag{C4}$$

The prefactor depending on c is a pure phase. Therefore

$$|\langle x| U |c + V\rangle|^2 = |\langle x| U |V\rangle|^2, \qquad \forall x, c, V, \tag{C5}$$

which is exactly the shift-invariance condition. □

Theorem (Shift invariance of fixed-phase HP-1)**.** *The fixed-phase HP-1 circuit is shift-invariant for HSP sampling over $\mathbb{Z}_{2^n}$.*

Proof. Let

$$\Lambda_1 = \{1, 3, 5, \ldots\}, \qquad \Lambda_2 = \{2, 4, 6, \ldots\}, \tag{C6}$$

and define the fixed-phase HP-1 circuit by

$$U_{\mathrm{fHP1}} = \left(\bigotimes_{j\in\Lambda_2} H_j\right)\left(\prod_{i\in\Lambda_1}\prod_{j\in\Lambda_2} \mathrm{CP}_{ij}(\theta_{ij})\right)\left(\bigotimes_{i\in\Lambda_1} H_i\right), \qquad \theta_{ij} = \frac{2\pi}{2^{|i-j|}}. \tag{C7}$$

For input and output bit strings y and x, the first Hadamard layer contributes the Walsh phase $(-1)^{x_i y_i}$ on qubits $i \in \Lambda_1$. The controlled-phase block then contributes $\exp\Big[\mathrm{i}\sum_{i\in\Lambda_1}\sum_{j\in\Lambda_2}\theta_{ij}x_i y_j\Big]$, because qubits in Λ_2 have not yet been acted on by Hadamards. The final Hadamard layer contributes $(-1)^{x_j y_j}$ on qubits $j \in \Lambda_2$. Hence

$$\langle x|\, U_{\mathrm{fHP1}}\, |y\rangle = \frac{1}{\sqrt{2^n}} \exp\left[\mathrm{i}\pi\sum_{k=1}^{n} x_k y_k + \mathrm{i}\sum_{i\in\Lambda_1}\sum_{j\in\Lambda_2}\theta_{ij}x_i y_j\right]. \tag{C8}$$

This is precisely the bilinear phase form in Lemma 1, with real coefficients

$$A_{ij} = \pi\delta_{ij} + \begin{cases}\theta_{ij}, & i\in\Lambda_1,\ j\in\Lambda_2,\\ 0, & \text{otherwise.}\end{cases} \tag{C9}$$

Lemma 1 therefore implies that U_{fHP1} is shift-invariant over $\mathbb{Z}_{2^n}$. □

D. Sampling calculation and recovery guarantee for the HP $-$ 0 case

We now give the calculation underlying the HP-0 discussion in the main text. Consider $G \simeq \mathbb{Z}_{2^n}$ represented by n computational-basis qubits. Because $\mathbb{Z}_{2^n}$ is a cyclic group of order 2^n, any of its subgroups must be generated by a divisor of 2^n. Thus, the hidden subgroup is necessarily of the form $V = \langle 2^p\rangle$ for some integer $0 \le p \le n$. Since every element $h \in V$ is a multiple of 2^p, its n-bit binary representation naturally ends with p zeros. Under this standard convention, an element h takes the form

$$h = h_1 h_2 \cdots h_{n-p} 0 \cdots 0. \tag{D1}$$

The subgroup generated by 2^p is

$$V = \{q2^p : q = 0, \ldots, 2^{n-p}-1\}, \tag{D2}$$

so $|V| = 2^{n-p}$. Since adding an element of V can vary the first $n-p$ bits, each coset has a representative of the form

$$c = 0\cdots 0 c_{n-p+1}\cdots c_n. \tag{D3}$$

Thus a normalized coset state can be written as

$$|c+V\rangle = \frac{1}{\sqrt{2^{n-p}}} \sum_{h_1,\ldots,h_{n-p}\in\{0,1\}} |h_1\cdots h_{n-p} c_{n-p+1}\cdots c_n\rangle. \tag{D4}$$

For an n-bit string x, applying Hadamard gates to all n qubits gives

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{k\in\{0,1\}^n} (-1)^{k\cdot x}|k\rangle, \tag{D5}$$

where $k\cdot x = \sum_{i=1}^n k_i x_i$ denotes the standard bitwise inner product of the binary strings k and x. Applying this to Eq. (D4) gives

$$H^{\otimes n}|c+V\rangle = \frac{1}{\sqrt{2^{n-p}}}\frac{1}{\sqrt{2^n}} \sum_{k\in\{0,1\}^n}\sum_{h_1,\ldots,h_{n-p}\in\{0,1\}} (-1)^{\sum_{i=1}^{n-p} k_i h_i + \sum_{i=n-p+1}^{n} k_i c_i}|k\rangle$$

Algorithm 1: HP $-$ 0 Sampling for $V \leq \mathbb{Z}_{2^n}$

Input: Oracle access to an HSP instance over $\mathbb{Z}_{2^n}$, sample count m
Output: Hidden subgroup $V = \langle 2^p \rangle$

Initialize an empty list of samples $\mathcal{B}$;
for $i = 1$ **to** m **do**
  Prepare the uniform superposition $\frac{1}{\sqrt{2^n}} \sum_{g \in \mathbb{Z}_{2^n}} |g\rangle |0\rangle$;
  Query the oracle (measuring the second register to obtain a coset state $|c_i + V\rangle$ is optional);
  Apply $H^{\otimes n}$ to the first register;
  Measure the first register and record the bit string $b^{(i)} \in \{0,1\}^n$;
  Append $b^{(i)}$ to $\mathcal{B}$;
end
Find the minimum index j^* (where $j = 1$ is the most significant bit) such that $b^{(i)}_{j^*} = 1$ for some sample in $\mathcal{B}$;
if *such a* j^* *exists* **then**
  Let $p = n - j^* + 1$;
else
  Let $p = 0$;
end
return $V = \langle 2^p \rangle$;

$$= \frac{1}{\sqrt{2^{n-p}}} \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{\sum_{i=n-p+1}^{n} k_i c_i} \prod_{i=1}^{n-p} \left(\sum_{h_i \in \{0,1\}} (-1)^{k_i h_i} \right) |k\rangle . \tag{D6}$$

The inner product factor is zero unless $k_1 = \cdots = k_{n-p} = 0$, in which case it equals 2^{n-p}. Therefore

$$H^{\otimes n} |c + V\rangle = \frac{1}{\sqrt{2^p}} \sum_{k_{n-p+1}, \ldots, k_n \in \{0,1\}} (-1)^{\sum_{i=n-p+1}^{n} k_i c_i} |0\rangle^{\otimes(n-p)} |k_{n-p+1} \cdots k_n\rangle . \tag{D7}$$

The coset representative c is eliminated in Eq. (D7). Consequently, the measurement distribution is independent of c. The first $n-p$ bits are always zero, while the remaining p bits are uniformly random.

This immediately gives a simple reconstruction procedure for the idealized $\mathbb{Z}_{2^n}$ HSP. Each application of the oracle-and-measurement routine prepares a random coset state, and Eq. (D7) shows that subsequently applying $H^{\otimes n}$ converts that coset state into a sample whose first $n-p$ bits are deterministically zero while the remaining p bits are unbiased. The resulting sampling-and-recovery routine is summarized in Algorithm 1.

We note that the physical measurement of the second register in Algorithm 1 is not strictly necessary. By the principle of deferred measurement, tracing out the second register leaves the first register in a mixed state of all possible cosets. Since the measurement distribution is strictly independent of the coset representative c, this yields identical measurement statistics. This device-friendly feature avoids the need for mid-circuit measurements, which is highly advantageous for implementations on near-term quantum hardware.

Algorithm 1 is correct because, under the bit-ordering convention above, the free positions occur exactly in the final p bits. Equivalently, the measurement data have the form

$$0 \cdots 0 * \cdots *, \tag{D8}$$

where each star denotes an unbiased bit. Therefore, the classical post-processing only needs to identify the leftmost bit position that ever evaluates to 1 across all m samples. This single position j^* uniquely determines p, and hence determines the subgroup $V = \langle 2^p \rangle$.

The sample complexity is logarithmic in the number of qubits. Under this leftmost-1 decoding strategy, the algorithm only fails if the most significant free bit (at position $n-p+1$) is measured as 0 in all m independent samples. Because this specific bit is uniformly random, the probability of it being zero m times in a row is exactly 2^{-m}. Thus $m \geq \lceil \log_2(1/\epsilon) \rceil$ samples are sufficient for Algorithm 1 to perfectly identify the hidden subgroup V with probability at least $1-\epsilon$. This tight bound elegantly removes the union-bound dependence on n, yielding an optimized sample complexity.

E. QFT Fisher information for subgroup-generator estimation

To benchmark the proposed circuit family, we treat the hidden subgroup generator r as the parameter to be inferred from the measurement distribution produced by the standard QFT. Consider the truncated periodic state

$$\begin{aligned} R &:= \left\lfloor \frac{2^n - 1}{r} \right\rfloor + 1 && \text{number of supported multiples of } r, \\ |\psi_r\rangle &:= \frac{1}{\sqrt{R}} \sum_{q=0}^{R-1} |qr\rangle && \text{normalized truncated periodic state.} \end{aligned} \tag{E1}$$

Thus $q = 0, \ldots, R-1$ labels precisely the integers qr contained in the 2^n-dimensional register. For the 2^n-point QFT,

$$|j\rangle \mapsto \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi \mathrm{i} jk/2^n} |k\rangle, \tag{E2}$$

and therefore

$$|\psi_r\rangle \mapsto \frac{1}{\sqrt{R2^n}} \sum_{k=0}^{2^n-1} \left(\sum_{q=0}^{R-1} e^{2\pi \mathrm{i} qrk/2^n} \right) |k\rangle. \tag{E3}$$

The corresponding measurement probability is

$$\Pr(k|r) = \frac{1}{R2^n} \left| \sum_{q=0}^{R-1} e^{2\pi \mathrm{i} qrk/2^n} \right|^2 = \frac{1}{R2^n} \left(\frac{\sin(\pi Rrk/2^n)}{\sin(\pi rk/2^n)} \right)^2. \tag{E4}$$

In the continuum approximation, the Fisher-information calculation reduces to a Dirichlet-kernel moment estimate. The result needed for the main-text comparison is summarized in the following lemma.

Lemma 3 (QFT Fisher-information)**.** *In the regime $1 \ll r \ll 2^n$, with $R \approx 2^n/r$, the QFT Fisher information has exponential rate $2\ln 2$ in n for fixed or polynomially bounded r.*

Proof. In the regime $r \ll 2^n$, introduce the continuum measurement coordinate and its probability density,

$$\begin{aligned} x &:= \frac{\pi k}{2^n} \in [0, \pi] && \text{continuum coordinate associated with outcome } k, \\ f(x, r) &:= \frac{1}{\pi R} \left(\frac{\sin(Rrx)}{\sin(rx)} \right)^2 && \text{continuum approximation to the QFT distribution.} \end{aligned} \tag{E5}$$

The QFT Fisher information with respect to the generator r is

$$\mathrm{DFI}_{\mathrm{QFT}}(r, n) = \int_0^\pi f(x, r) \left[\frac{\partial}{\partial r} \ln f(x, r) \right]^2 dx, \tag{E6}$$

with score function

$$s_r(x) := \frac{\partial}{\partial r} \ln f(x, r) = 2Rx \cot(Rrx) - 2x \cot(rx), \qquad s_r(x) = \text{score for the generator } r. \tag{E7}$$

Define

$$D_R(u) := \frac{\sin(Ru)}{\sin u}, \qquad D_R(u) = \text{order-}R \text{ Dirichlet kernel.} \tag{E8}$$

Then $f(x, r) = D_R(rx)^2/(\pi R)$, and the score function implies

$$f(x, r) \left[\frac{\partial}{\partial r} \ln f(x, r) \right]^2 = \frac{4x^2}{\pi R} \left[D_R'(rx) \right]^2. \tag{E9}$$

Thus the Fisher-information integral is reduced to the square of a finite trigonometric polynomial. Once that polynomial is expanded into cosine modes, the remaining calculation involves only the moments of x^2 against those modes.

We begin by expressing the Dirichlet kernel $D_R(u)$ in its exponential sum form. By shifting the summation index to center the phases, it can be written as a symmetric finite series:

$$D_R(u) = \frac{\sin(Ru)}{\sin u} = \sum_{m=0}^{R-1} e^{\mathrm{i}(R-1-2m)u}, \tag{E10}$$

Differentiating this series term-by-term with respect to u yields:

$$D'_R(u) = \mathrm{i} \sum_{m=0}^{R-1} (R-1-2m) e^{\mathrm{i}(R-1-2m)u}. \tag{E11}$$

To compute the squared magnitude $[D'_R(u)]^2$, we multiply this sum by itself. By collecting terms that produce the same relative frequency difference, we can perfectly recast the product as an even cosine series:

$$[D'_R(u)]^2 = A_0 + 2 \sum_{j=1}^{R-1} A_j \cos(2ju), \tag{E12}$$

where A_j denotes the cosine-series coefficient at frequency $2j$. These coefficients are obtained by evaluating the convolution sum over amplitudes with frequency offset j. To evaluate this sum explicitly, we expand the summand as a quadratic in m and apply the standard summation formulas for $\sum m$ and $\sum m^2$:

$$\begin{aligned} A_j &= -\sum_{m=0}^{R-1-j} (R-1-2m)(2j-R+1+2m) \\ &= \sum_{m=0}^{R-1-j} \left[4m^2 - 4(R-1-j)m + (R-1-j)^2 - j^2\right] \\ &= \frac{R-j}{3}\left[(R-j-1)^2 + 2(R-j-1) - 3j^2\right] \\ &= \frac{(R-j)(R^2-2Rj-2j^2-1)}{3}, \qquad j = 0, \dots, R-1. \end{aligned} \tag{E13}$$

In particular, setting $j = 0$ isolates the non-oscillatory constant term:

$$A_0 = \frac{R(R^2-1)}{3}. \tag{E14}$$

Recall from our earlier derivation of the score function that the Fisher information integral takes the form:

$$\mathrm{DFI}_{\mathrm{QFT}}(r,n) = \frac{4}{\pi R} \int_0^\pi x^2 [D'_R(rx)]^2 \, dx, \tag{E15}$$

Substituting our exact cosine expansion of $[D'_R(rx)]^2$ into this integral distributes the integration across the individual frequency modes:

$$\mathrm{DFI}_{\mathrm{QFT}}(r,n) = \frac{4}{\pi R} \left[A_0 \int_0^\pi x^2 \, dx + 2 \sum_{j=1}^{R-1} A_j \int_0^\pi x^2 \cos(2jrx) \, dx \right]. \tag{E16}$$

The problem is now reduced to evaluating two elementary definite integrals over $[0, \pi]$. Define the non-oscillatory and oscillatory moments by

$$\begin{aligned} M_0 &:= \int_0^\pi x^2 \, dx = \frac{\pi^3}{3} && \text{non-oscillatory second moment,} \\ M_j(r) &:= \int_0^\pi x^2 \cos(2jrx) \, dx && \text{oscillatory second moment at mode } j \\ &= \frac{4\pi jr \cos(2\pi jr) + (4\pi^2 j^2 r^2 - 2)\sin(2\pi jr)}{(2jr)^3}, \qquad j \geq 1. \end{aligned} \tag{E17}$$

Inserting these evaluated moments, along with the explicit expressions for A_0 and A_j, back into the expanded Fisher information equation gives:

$$\begin{aligned}\mathrm{DFI}_{\mathrm{QFT}}(r,n) &= \frac{4}{\pi R}\left[\frac{R(R^2-1)}{3}\cdot\frac{\pi^3}{3}+\frac{2}{3}\sum_{j=1}^{R-1}(R-j)(R^2-2Rj-2j^2-1)\,M_j(r)\right] \\ &= \frac{4\pi^2}{9}(R^2-1)+\frac{8}{3\pi R}\sum_{j=1}^{R-1}(R-j)(R^2-2Rj-2j^2-1)\,M_j(r),\end{aligned} \tag{E18}$$

which is the stated exact formula.

Finally, we analyze the asymptotic behavior in the standard regime of interest, $1 \ll r \ll R$. In this limit, the hidden parameter is large enough to suppress oscillations, yet small enough that its multiples are heavily supported in the register. The explicit form of the oscillatory moments shows that $M_j(r) = \mathcal{O}(r^{-2})$ for any fixed mode j. Since the leading constant term scales as $\mathcal{O}(R^2)$ and the summation over the modes scales proportionally to $\mathcal{O}(R^2r^{-2})$, the relative magnitude of the oscillatory corrections is heavily suppressed by a factor of $1/r^2$. Because these high-frequency corrections are strictly subleading in this large-r regime, the behavior is overwhelmingly governed by the constant contribution. Keeping only this non-oscillatory term and substituting $R \approx 2^n/r$ therefore gives:

$$\mathrm{DFI}_{\mathrm{QFT}}(r,n) \approx \frac{4\pi^2}{9}\left(\frac{2^{2n}}{r^2}-1\right). \tag{E19}$$

This proves the lemma. □

Having established the QFT Fisher-information benchmark, we now turn to the corresponding HP-circuit measurement distribution. To study the DFI of HP circuits, the first step is to express their output probabilities in a form that separates the phase-dependent circuit coefficients from the period- dependent interference terms.

For readability, write the local phase sum seen by output bit j as

$$\phi_j(x) := \sum_{i\neq j}\tilde{\theta}_{ij}x_i, \qquad \phi_j(x) = \text{local phase sum at output bit } j, \qquad j = 1,\dots,n. \tag{E20}$$

For a mask $\Delta \in \{0,1\}^n$, define its Hamming weight and the single-bit masks by

$$\begin{aligned} |\Delta| &:= \sum_{j=1}^{n}\Delta_j && \text{number of flipped Walsh factors,} \\ 0 &:= (0,\dots,0) && \text{zero-flip mask,} \\ e_m &:= (0,\dots,0,1,0,\dots,0) && \text{single-flip mask at bit } m, \\ (x\oplus\Delta)_j &:= x_j+\Delta_j \pmod 2 && \text{bitwise XOR of } x \text{ and } \Delta. \end{aligned} \tag{E21}$$

where the only nonzero entry of e_m is at bit m. The phase-dependent Walsh coefficient associated with Δ is

$$B_\Delta(x) = \prod_{j=1}^{n}\exp\left(\frac{\mathrm{i}}{2}\phi_j(x)\right)\left[\cos\left(\frac{\phi_j(x)}{2}\right)\right]^{1-\Delta_j}\left[-\mathrm{i}\sin\left(\frac{\phi_j(x)}{2}\right)\right]^{\Delta_j}. \tag{E22}$$

The low-weight symbols used below are therefore

$$\begin{aligned} B_0(x) &:= B_{\Delta=0}(x) && \text{zero-flip Walsh amplitude,} \\ |B_0(x)|^2 &= |B_{\Delta=0}(x)|^2 && \text{zero-flip Walsh weight,} \\ B_{e_m}(x) &:= B_{\Delta=e_m}(x) && \text{single-flip Walsh amplitude at bit } m. \end{aligned} \tag{E23}$$

Thus $B_0(x)$ is an amplitude, whereas its squared modulus $|B_0(x)|^2$ is its zero-flip Walsh weight. For the periodic state

$$|V_r\rangle = \frac{1}{\sqrt{R}}\sum_{q=0}^{R-1}|qr\rangle\,, \qquad V_r = \{qr : q = 0,1,\dots,R-1\}, \tag{E24}$$

with each integer qr represented as an n-bit computational-basis string, set

$$W_r(k) := \sum_{q=0}^{R-1} (-1)^{\sum_{j=1}^{n} k_j (qr)_j} \qquad \text{Walsh character sum over the periodic support,} \tag{E25}$$
$$k_j = \text{the } j\text{-th bit of } k, \qquad (qr)_j = \text{the } j\text{-th bit of } qr.$$

The sign factor $(-1)^{\sum_j k_j y_j}$ is the standard Walsh kernel on the Boolean group $(\mathbb{Z}_2)^n$ [38, 39]; hence $W_r(k)$ is a Walsh character sum over the sampled periodic support.

Lemma 4 (Walsh form of the HP measurement probabilities)**.** *With the definitions above, the HP measurement probability is*

$$\Pr(x \mid r) = \frac{1}{R2^n} \left| \sum_{\Delta \in \{0,1\}^n} B_\Delta(x) W_r(x \oplus \Delta) \right|^2 . \tag{E26}$$

Thus the phase-dependent coefficients and period-dependent interference terms are separated explicitly.

Proof. To prove the probability formula, first derive the HP circuit matrix element,

$$\langle x | \, U_{\mathrm{HP}}^{(L)}(\boldsymbol{\theta}) \, | y \rangle = \frac{1}{\sqrt{2^n}} \exp\left(\mathrm{i} \sum_{i,j} \tilde{\theta}_{ij} x_i y_j \right). \tag{E27}$$

Since $\tilde{\theta}_{jj} = \pi$, separating the diagonal and off-diagonal terms gives

$$\langle x | \, U_{\mathrm{HP}}^{(L)}(\boldsymbol{\theta}) \, | y \rangle = \frac{(-1)^{\sum_{j=1}^{n} x_j y_j}}{\sqrt{2^n}} \prod_{j=1}^{n} \exp\left(\mathrm{i} y_j \phi_j(x) \right). \tag{E28}$$

For each Boolean input bit $y_j \in \{0,1\}$,

$$y_j = \frac{1 - (-1)^{y_j}}{2}. \tag{E29}$$

Therefore

$$\exp(\mathrm{i} y_j \phi_j(x)) = \exp\left(\frac{\mathrm{i}}{2} \phi_j(x) \right) \left[\cos\left(\frac{\phi_j(x)}{2} \right) - \mathrm{i} (-1)^{y_j} \sin\left(\frac{\phi_j(x)}{2} \right) \right]. \tag{E30}$$

Multiplying over j and collecting terms by the mask Δ gives

$$\exp\left(\mathrm{i} \sum_{j=1}^{n} y_j \phi_j(x) \right) = \sum_{\Delta \in \{0,1\}^n} B_\Delta(x) (-1)^{\sum_{j=1}^{n} \Delta_j y_j}. \tag{E31}$$

Substituting this expansion into the separated matrix element gives

$$\langle x | \, U_{\mathrm{HP}}^{(L)}(\boldsymbol{\theta}) \, | y \rangle = \frac{1}{\sqrt{2^n}} \sum_{\Delta \in \{0,1\}^n} B_\Delta(x) (-1)^{\sum_{j=1}^{n} (x_j \oplus \Delta_j) y_j}. \tag{E32}$$

Substituting $y = qr$ and summing over the uniformly weighted components of $|V_r\rangle$ yields

$$\langle x | \, U_{\mathrm{HP}}^{(L)}(\boldsymbol{\theta}) \, | V_r \rangle = \frac{1}{\sqrt{R2^n}} \sum_{\Delta \in \{0,1\}^n} B_\Delta(x) W_r(x \oplus \Delta). \tag{E33}$$

Taking the squared modulus proves the lemma. □

The Walsh expansion above gives a useful way to interpret the numerical DFI growth, but it does not by itself prove a universal asymptotic law for all HP phase choices. In particular, the higher-weight Walsh terms in Lemma 4 need not be uniformly negligible for an arbitrary circuit. The following result should therefore be read as a conditional scaling model: it states what follows when the observed numerical regime is dominated by the coherent low-weight component.

Theorem 5 (Conditional HP Fisher-information scaling model)**.** *We use the following notation in the assumptions:*

$$\begin{aligned}
\Delta_r \Pr(x \mid r) &:= \Pr(x \mid r+1) - \Pr(x \mid r) && \textit{finite difference in the period generator } r, \\
\Pr_{\mathrm{bg}}(x \mid r) &:= \textit{residual HP probability} && \textit{background from Walsh masks with } |\Delta| \geq 1, \\
\gamma &:= \textit{typical large-}n\textit{ limit of } \gamma_n(x) && \textit{coherent attenuation rate per qubit}, \\
A_n &:= \exp[-\gamma n + o(n)] && \textit{coherent attenuation factor from } |B_0(x)|^2.
\end{aligned} \tag{E34}$$

Suppose that, on the output region contributing dominantly to the DFI sum, the following conditions hold: (i) the coherent zero-flip Walsh weight has a typical exponential rate $|B_0(x)|^2 = \exp[-\gamma n + o(n)]$*; (ii) the higher-weight Walsh terms can be collected into a background* $\Pr_{\mathrm{bg}}$ *whose finite difference in* r *is negligible in that region; and (iii) the coherent contribution is peak-dominant in the sense that* $A_n \Pr_{\mathrm{QFT}} \gg \Pr_{\mathrm{bg}}$*, with* $A_n = \exp[-\gamma n + o(n)]$*. Under these assumptions, the HP Fisher information obeys the scaling model*

$$\mathrm{DFI}_{\mathrm{HP}}(n) \approx \exp\left[(2\ln 2 - \gamma)n + o(n)\right]. \tag{E35}$$

Equivalently, the leading exponent is $k = 2\ln 2 - \gamma$*.*

Proof. The Walsh probability formula can be organized by the Hamming weight of Δ. The first two weight classes are

$$\begin{aligned}
|\Delta| = 0: &\quad B_0(x) W_r(x) && \text{zero-flip coherent Walsh term}, \\
|\Delta| = 1: &\quad B_{e_m}(x) W_r(x \oplus e_m), \quad m = 1, \dots, n && \text{single-flip shifted Walsh term}.
\end{aligned} \tag{E36}$$

For real phases, and assuming the relevant cosine factors in $B_0(x)$ are nonzero when tangent ratios are used,

$$B_0(x) = \prod_{j=1}^{n} \exp\left(\frac{\mathrm{i}}{2}\phi_j(x)\right) \cos\left(\frac{\phi_j(x)}{2}\right). \tag{E37}$$

$$\begin{aligned}
B_{e_m}(x) &= \exp\left(\frac{\mathrm{i}}{2}\sum_{j=1}^{n}\phi_j(x)\right)\left[-\mathrm{i}\sin\left(\frac{\phi_m(x)}{2}\right)\right]\prod_{j\neq m}\cos\left(\frac{\phi_j(x)}{2}\right) \\
&= B_0(x)\left[-\mathrm{i}\tan\left(\frac{\phi_m(x)}{2}\right)\right].
\end{aligned} \tag{E38}$$

For compactness, define the single-flip tangent ratio

$$T_m(x) := \tan\left(\frac{\phi_m(x)}{2}\right), \qquad T_m(x) = \text{relative amplitude factor for a flip at bit } m. \tag{E39}$$

Then $B_{e_m}(x) = -\mathrm{i}B_0(x)T_m(x)$. Factoring out $B_0(x)$, the exact HP probability can be written as the zero-flip and single-flip contribution plus a higher-weight remainder,

$$\Pr_{\mathrm{HP}}(x \mid r) = \frac{|B_0(x)|^2}{R 2^n}\left|W_r(x) - \mathrm{i}\sum_{m=1}^{n} T_m(x) W_r(x \oplus e_m) + \mathcal{R}_{>1}(x, r)\right|^2. \tag{E40}$$

Here the normalized higher-weight remainder is

$$\mathcal{R}_{>1}(x, r) := B_0(x)^{-1} \sum_{|\Delta|>1} B_\Delta(x) W_r(x \oplus \Delta). \tag{E41}$$

Because $W_r(k)$ and the explicit phase sums are real, the displayed zero-flip term and single-flip correction have zero linear interference:

$$2\,\mathrm{Re}\left[W_r(x)^*(-\mathrm{i})\sum_{m=1}^{n} T_m(x) W_r(x \oplus e_m)\right] = 0. \tag{E42}$$

The coherent zero-flip weight is therefore

$$|B_0(x)|^2 = \prod_{j=1}^{n} \cos^2\left(\frac{\phi_j(x)}{2}\right). \tag{E43}$$

This factor is the analogue of a Debye–Waller factor, the standard coherent-scattering attenuation factor associated with fluctuations [40]: it is the multiplicative attenuation of the coherent peak caused by the phase-sum fluctuations. Here the coherent peak is the zero-flip Walsh component, and its attenuation factor is exactly $|B_0(x)|^2$. We therefore define the Debye–Waller rate as the negative logarithm of this attenuation per qubit,

$$\gamma_n(x) := -\frac{1}{n}\ln|B_0(x)|^2 = -\frac{1}{n}\sum_{j=1}^{n}\ln\cos^2\left(\frac{\phi_j(x)}{2}\right), \qquad \gamma_n(x) = \text{finite-size coherent-loss rate per qubit.} \tag{E44}$$

Equivalently, $|B_0(x)|^2 = \exp[-n\gamma_n(x)]$, so $\gamma_n(x)$ measures the exponential loss rate of coherent Walsh peak weight. For small values of these phase sums, $\gamma_n(x) = \frac{1}{4n}\sum_{j=1}^{n}\phi_j(x)^2 + O\Big(n^{-1}\sum_{j=1}^{n}|\phi_j(x)|^4\Big)$.

The zero-flip Walsh term is not identical to the QFT Dirichlet kernel. Instead, it is used here as the coherent periodic interference component in a phenomenological model for the observed HP data. When $\gamma_n(x)$ converges to a typical value γ on the DFI-dominant output region, the coherent peak has weight $A_n = \exp[-\gamma n + o(n)]$. In that same region the conditional peak-dominance model is

$$\Pr_{\mathrm{HP}}(x \mid r) \approx A_n \Pr_{\mathrm{QFT}}(x \mid r) + \Pr_{\mathrm{bg}}(x \mid r), \qquad A_n = \exp[-\gamma n + o(n)], \tag{E45}$$

where $\Pr_{\mathrm{bg}}$ absorbs the single-flip squared terms, the higher-weight Walsh remainder, and their residual interference terms. The theorem assumes that this background is subdominant for the DFI exponent. Without those assumptions, the exact Walsh formula remains valid, but the exponential rate stated here is not implied.

In the DFI-dominant region, these assumptions imply $\Delta_r \Pr_{\mathrm{HP}} \approx A_n \Delta_r \Pr_{\mathrm{QFT}}$ and $\Pr_{\mathrm{HP}} \approx A_n \Pr_{\mathrm{QFT}}$. Substituting these relations into

$$\mathrm{DFI}(r) = \sum_x \frac{(\Delta_r \Pr(x \mid r))^2}{\Pr(x \mid r)} \tag{E46}$$

gives $\mathrm{DFI}_{\mathrm{HP}} \approx A_n \mathrm{DFI}_{\mathrm{QFT}}$. Since $A_n = \exp[-\gamma n + o(n)]$ and $\mathrm{DFI}_{\mathrm{QFT}} \approx \exp[2\ln 2\, n + o(n)]$, the stated scaling follows. □

F. Log-linear growth fit for the numerical Fisher-information scan

The fixed-r QFT benchmark derived above clarifies the dominant scaling mechanism, but Fig. 2 probes a slightly different quantity. There, the vertical axis is the minimum discrete Fisher information over the period window used in the numerical scan. To ensure the asymptotic condition $1 \ll r \ll 2^n$ is strictly maintained as the system scales, the numerical scan uses the polynomial period window $2 \le r \le n^2$. The relevant QFT comparator is therefore the worst case within this dynamically scaled window rather than a single fixed value of r.

The leading term in Lemma 3 is monotone in R, and in the sampled regime $R \approx 2^n/r$. Consequently, the smallest QFT Fisher information inside the scan window is attained near the largest sampled period. Substituting $r_{\max}(n) = n^2$ yields the window-matched estimate

$$\mathrm{DFI}_{\mathrm{QFT,min}}(n) \approx \frac{4\pi^2}{9}\left(\left\lfloor\frac{2^n}{n^2}\right\rfloor^2 - 1\right). \tag{F1}$$

If one keeps only the dominant exponential factor, the logarithmic growth becomes

$$\ln \mathrm{DFI}_{\mathrm{QFT,min}}(n) = 2n\ln 2 - 4\ln n + \mathcal{O}(1). \tag{F2}$$

Thus, the asymptotic QFT exponent under this polynomial period scaling approaches $k_{\mathrm{QFT},\infty} = 2\ln 2$, with a finite-window reduction from the $-4\ln n$ term.

This observation suggests a simple diagnostic for the numerical data in Fig. 2. If the plotted curves retain a stable exponential dependence on the qubit count over the sampled window, then the transformed data should be well approximated by

$$\ln \mathrm{DFI}_{\min}(n) = kn + b. \tag{F3}$$

We therefore perform ordinary least-squares fits to the log-transformed mean values of each curve shown in Fig. 2. Table I summarizes the fitted exponents; we do not repeat the transformed plot because it carries the same point set as Fig. 2.

Circuit	qubits	k	95% CI	R^2
HP-1 fixed	7–18	0.368	[0.324, 0.413]	0.972
HP-1 shared optimized	7–18	0.379	[0.354, 0.403]	0.992
HP-L random	7–18	0.398	[0.383, 0.414]	0.997
HP-L random with phase	7–18	0.352	[0.336, 0.367]	0.996

TABLE I. Effective exponents obtained from the log-linear model $\ln \mathrm{DFI}_{\min}(n) = kn + b$ applied to the numerical Fisher-information curves in Fig. 2.

The fitted lines account for most of the log-scale variance across the sampled window. The coefficient of determination (R^2) ranges from 0.972 to 0.997, the log-space root-mean-square residual lies between 0.075 and 0.217, and every fitted slope (k) is significantly positive with a p-value $\leq 4.6 \times 10^{-9}$. In that sense, a linear regression on $\ln I_{\min}$ is justified as a finite-window descriptive model for these curves. At the same time, the fitted slope k should be interpreted as an effective exponent on the restricted range $n = 7, \ldots, 18$ (where n is the number of qubits), not as a claim about the ultimate asymptotic law.

The comparison with the QFT benchmark is unambiguous. Even after matching the period window used in the numerical scan, the QFT reference fit remains much steeper, with $k_{\mathrm{QFT,win}} = 1.089$ and $R^2 = 0.999$. The empirical circuit families therefore grow substantially more slowly: their fitted exponents occupy 32%–37% of the window-matched QFT slope, and 25%–29% of the asymptotic QFT exponent $2\ln 2$. Among the plotted families, the random-layer HP circuit has the largest point estimate, near $k = 0.398$, followed by the shared optimized HP-1 circuit, near $k = 0.379$.

This comparison sharpens the interpretation of Fig. 2. The HP family circuits retain a clear exponential-in-n growth trend in the $\mathrm{DFI}_{\min}$ over the sampled window, which is why the log-linear fit works well, but they do not reproduce the QFT growth rate. Their advantage in the present setting is therefore not that they match the full QFT benchmark asymptotically, but that they retain a nontrivial and practically usable fraction of that information while remaining much shallower.

G. Noise-robustness experiment details

We evaluate noise robustness using the previously trained 9- and 10-qubit HP-1 plus neural-decoder pipelines from the period-recovery study. For each system size, we use a measurement budget of $1024n^2$ samples per instance, giving 82944 measurements for $n = 9$ and 102400 for $n = 10$. The candidate periods are $r \in \{9, \ldots, 80\}$ for 9 qubits and $r \in \{10, \ldots, 99\}$ for 10 qubits. Probability distributions are generated from the exact subgroup support of the trained HP-1 circuit, and the neural decoder checkpoint is selected using noiseless validation accuracy. The selected checkpoints occur at epoch 3 with validation top-1 accuracy 0.9959 for 9 qubits, and at epoch 10 with validation top-1 accuracy 1.0000 for 10 qubits.

The evaluation set holds out 3 shifts for each candidate period and generates 8 independent noisy redraws for each held-out shift. This yields $72 \times 3 \times 8 = 1728$ test examples for 9 qubits and $90 \times 3 \times 8 = 2160$ test examples for 10 qubits. To probe robustness after the shallow quantum transformation, we insert a single global depolarizing channel after the full HP-1 circuit and before measurement. This is the 2^n-dimensional version of the standard depolarizing model, where the ideal output is mixed with the maximally mixed state; global depolarizing noise is also used in circuit-level noise analyses [41, 42]. At the probability level, this is implemented as

$$\widetilde{\Pr}_\eta(x|r,s) = (1-\eta)\Pr(x|r,s) + \eta\frac{1}{2^n}, \tag{G1}$$

where $\Pr(x|r, s)$ is the ideal output distribution for period r and shift s, and η is the depolarizing strength. For each noisy distribution $\widetilde{\Pr}_\eta$, we resample $1024n^2$ outcomes and feed the resulting bit-string batch into the fixed neural decoder. The sweep uses the values

$$\eta \in \{1,\ 0.464,\ 0.215,\ 0.1,\ 0.0464,\ 0.0215,\ 0.01,\ 0.00464,\ 0.00215,\ 0.001\}. \tag{G2}$$

η	9-qubit accuracy	10-qubit accuracy
1.000	0.0139	0.0111
0.464	0.4913	0.8810
0.215	0.9948	1.0000
0.100	0.9954	1.0000
0.0464	0.9959	1.0000
0.0215	0.9942	1.0000
0.0100	0.9954	1.0000
0.00464	0.9959	1.0000
0.00215	0.9948	1.0000
0.00100	0.9948	1.0000

TABLE II. Held-out period-recovery accuracy under post-HP-1 global depolarizing noise.

Several features of Table II are worth noting. First, performance is essentially unchanged throughout the weak-to-moderate noise regime and remains near the noiseless baseline up to $\eta \approx 0.215$. Second, substantial degradation appears only once the noisy distribution is strongly mixed with the uniform distribution: at $\eta = 0.464$, the accuracy drops to 0.4913 for 9 qubits and 0.8810 for 10 qubits. Finally, when $\eta = 1$, the output distribution becomes exactly uniform and is therefore independent of both the period and the shift. In that limit, all subgroup information is erased, so the best possible behavior is chance-level guessing over the candidate periods. The observed accuracies 0.0139 and 0.0111 agree exactly with the corresponding chance levels $1/72$ and $1/90$, confirming that the numerical results are consistent with the depolarizing-noise model.